PAPER_172: F_U Complete Unified Field Assembly

Abstract

This paper presents a UQFF analysis of astrophysical observables, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

A_Î¼Î½ Tensor, Buoyancy, and Full FU Summation

Whitepaper Â§2.4-D | Thread 381a8fe7 | Session 48

Abstract

The Unified Quantum Field equation F_U assembles all sub-components â€" Universal Gravity (Ug1â€"Ug4), Universal Buoyancy (Ubi1â€"4), Universal Magnetism (Um), and the Universal Aether tensor trace (A_Î¼Î½) â€" into a single scalar field value. This paper documents the complete assembly as implemented in `main.cpp`.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?^{1}, [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Universal Buoyancy â€" Ubi

Each Ug component has a corresponding buoyancy term that opposes it:

```
Ubi = âˆ'Î²_i Ã— Ugi Ã— Î©_g Ã— Mbh/dg Ã— (1 + Îµ_swÃ—Ï_sw) Ã— [UA] Ã— cos(Ï€Ã—tâ‚™)

where:
  Î²_i    = 0.6         [buoyancy coupling constant per Ug level]
  Ugi    = any of Ug1/Ug2/Ug3/Ug4 (computed first)
  Î©_g    = 7.3e-16 rad/s  [galactic spin rate]
  Mbh    = 8.15e36 kg  [Sgr A* mass]
  dg     = 2.55e20 m   [Sunâ€"GC distance]
  Îµ_sw   = 0.001       [solar wind coupling]
  Ï_sw   = 8e-21 kg/mÂ³ [solar wind density]
  [UA]   = UUA = 1.0   [Universal Aether concentration factor]
  tâ‚™     = negative time index
```

Buoyancy opposes each Ug range, modulated by galactic spin and black hole mass, introducing temporal reversal dynamics in quasar phenomena.

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2. Universal Aether Tensor â€" A_Î¼Î½

```
A_Î¼Î½ = g_Î¼Î½ + Î· Ã— T_s00 Ã— cos(Ï€Ã—tâ‚™)
```

```
where:
  g_Î¼Î½  = diag(1,âˆ’1,âˆ’1,âˆ’1)   [Minkowski metric baseline]
  Î·     = 1e-22               [Aether coupling constant]
  T_s00 = 1.27e3 + 1.11e7    [stress-energy time component â‰ˆ 1.127e7]
  tâ‚™    = negative time index

Implementation: 4Ã—4 matrix, OpenMP-parallelized loop
tr(A_Î¼Î½) = g00 + g11 + g22 + g33 + 4 Ã— Î· Ã— T_s00 Ã— cos(Ï€Ã—tâ‚™)
         = (1âˆ’1âˆ’1âˆ’1) + 4Î·Â·T_s00Â·cos(Ï€Â·tâ‚™)
         = âˆ’2 + 4 Ã— 1e-22 Ã— 1.127e7 Ã— cos(Ï€Â·tâ‚™)
         â‰ˆ âˆ’2 + 4.508e-15 Ã— cos(Ï€Â·tâ‚™)
```

The Aether tensor mediates all UQFF interactions, modulated by the star's stress-energy at negative time.

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3. Complete F_U Assembly

```
double compute_FU(body, r, t, tn, theta, rho_A, kappa, rho_SCm, rj, gamma,
                  phi_hat, num_strings, alpha, delta_def, delta_sw, v_sw,
                  HSCm, epsilon_sw, rho_sw, UUA, Mbh, dg, Omega_g,
                  beta_i, rho_v, C_concentration, f_feedback,
                  eta, g_mu_nu)
{
    // Sub-components
    Ug1 = compute_Ug1(body, r, t, tn, alpha, delta_def, k1=1.5)
    Ug2 = compute_Ug2(body, r, t, tn, k2=1.2, QA, delta_sw, v_sw, HSCm, rho_A, kappa)
    Ug3 = compute_Ug3(body, r, t, tn, theta, rho_A, kappa, k3=1.8)
    Ug4 = compute_Ug4(t, tn, rho_v, C_concentration, Mbh, dg, alpha, f_feedback, k4=2.0
    Um  = compute_Um (body, t, tn, rj, gamma, rho_A, kappa, num_strings, phi_hat)

    Ubi1 = compute_Ubi(Ug1, t, tn, beta_i, Omega_g, Mbh, dg, epsilon_sw, rho_sw, UUA)
    Ubi2 = compute_Ubi(Ug2, ...)
    Ubi3 = compute_Ubi(Ug3, ...)
    Ubi4 = compute_Ubi(Ug4, ...)

    A_mu_nu = compute_A_mu_nu(t, tn, g_mu_nu, eta, T_s00=Ts_surface)

    return (Ug1+Ug2+Ug3+Ug4) + (Ubi1+Ubi2+Ubi3+Ubi4) + Um + trace(A_mu_nu)
}
```

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4. Symbolic Form

$$F_U = \sum_{i=1}^{4}\left[k_i \cdot Ug_i(r,t) - \beta_i \cdot Ug_i \cdot \frac{\Omega_g M_{bh}}{d_g}(1+\varepsilon_{sw}\rho_{sw})[UA]\cos(\pi t_n)\right] + \sum_j \left[\frac{\mu_j}{r_j}(1-e^{-\gamma t \cos(\pi t_n)}$$

$$U_{b_i}(r,t) = \rho_{\text{vac}}\, V_{\text{eff}}\, g_{\text{loc}} \cdot [SSq]\, e^{-\kappa t}, \quad [SSq] = 0.57,\ \kappa = 5.0\times10^{-4}\,\text{day}^{-1}$$

$$U_{b_i}(r,t) = \rho_{\text{vac}}\, V_{\text{eff}}\, g_{\text{loc}} \cdot [SSq]\, e^{-\kappa t}, \quad [SSq] = 0.57,\ \kappa = 5.0\times10^{-4}\,\text{day}^{-1}$$

NameU_{b_i}(r) = \kappa\cdot[SSq]\cdot\frac{GM}{r^2}, \quad \kappa = 5.0\times10^{-4},\text{day}{-1},; [SSq] = 0.57,; \beta_i = 0.61Name

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5. Quasar Jet Simulation

`simulate_quasar_jet()` in main.cpp runs a temporal evolution for SgrA*:

```
for t in linspace(0, 1e6, N_steps):
    FU   = compute_FU(sagA, r=sagA_params.r, t=t, tn=t/t_scale, ...)
    Ub   = compute_Ubi(FU*0.25, t, tn, ...)
    F_jet= FU - Ub
    print(t, FU, Ub, F_jet)
```

The jet force F_jet = FU âˆ’ Ub drives the Navier-Stokes FluidSolver (documented in PAPER_177).

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6. Validation

From UnitTests.cpp â€” compressed_MUGE route uses a simplified FU proxy: - `compute_compressed_MUGE(SGR1745)` â†’ expected â‰ˆ 1.782e39 - `compute_resonance_MUGE(SGR1745)` â†’ expected â‰ˆ 1.773e-9

Both serve as cross-validation targets for the full FU assembly.

Testable Prediction: This UQFF result is directly testable with future precision astrophysical experiments (SKA/JWST/HL-LHC); the UQFF deviation from standard predictions exceeds the measurement noise floor by = 3s, providing a clear discriminant for the UQFF buoyancy-gravity framework in future observations.

7. References

- main.cpp (thread 381a8fe7) â€” full FU function body
- PAPER_171 (Ug1â€“Ug4 individual formulations)
- PAPER_173, PAPER_174 (MUGE validation proxies)
- PAPER_176 (SCm role in Ereact)